

China starts operation on world's largest single-dish radio telescope



FAST will explore the Universe at radio frequencies, detecting emission from hydrogen gas in the Milky Way and other distant galaxies, finding faint galactic and extragalactic pulsars, and searching for potential radio signals from extraterrestrials (Image courtesy: <http://fast.bao.ac.cn>)

China has finalised the world's biggest radio telescope, whose 503m (1650-feet)-wide dish will scan the heavens for signs of intelligent alien life, among other tasks.

With a dish the size of 30 football fields, Five-hundred-meter Aperture Spherical radio Telescope (FAST) is by far the largest single-aperture telescope in the world (though arrays that link up multiple radio dishes cover more ground). The previous record holder in the field is the 300m (1000-feet)-wide Arecibo Observatory in Puerto Rico.

FAST was built in China's Guizhou Province, more than 2000 kilometres southwest of Beijing. The US\$180-million facility should help scientists learn more about the Universe's early days, detect low-frequency gravitational waves and hunt for signals that may have been produced by distant alien civilisations.

Computer that can solve problems traditional computers cannot

Processing power of standard computers is likely to reach its maximum in the next 10 to 25 years. Even at this maximum power, traditional computers would not be able to handle a particular class of problems that involve combining variables to come up with many possible answers, and looking for the best solution. Combining optical and electronic technology, Stanford researchers have made a new type of computer that can solve problems that are a challenge for traditional computers.

This new type of computer, which blends optical and electrical processing, could get around this impending processing constraint and solve those problems. If it can be scaled up, this non-traditional computer could save costs by finding more optimal solutions to problems that have an incredibly high number of possible solutions.

Peter McMahon, post-doctoral scholar in applied physics and co-author of the paper, says, "This is a machine that is in a sense the first in its class, and the

idea is that it opens up a sub-field of research in the area of non-traditional computing machines."



Post-doctoral scholar Peter McMahon (left) and visiting researcher Alireza Marandi (right) examine a prototype of the new type of light based computer (Image courtesy: <http://news.stanford.edu>)